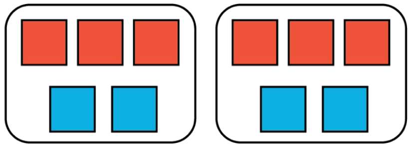
## 5.2.6 Wrap-Up: Ticket Out the Door

1. The representation shows the amount of red and blue paint that makes 2 batches of a particular shade of purple paint called lilac.

Cups of Red Paint

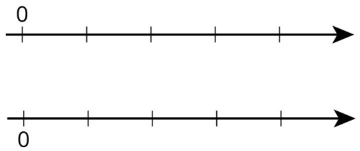
Cups of Blue Paint



   a. On the double number line, label the tick marks to represent amounts of red and blue paint used to make batches of this shade of purple paint.

Cups of Red Paint

Cups of Blue Paint




   b. How many batches are made with 12 cups of red paint?

   c. How much red paint should be mixed with 6 cups of blue paint to make lilac paint?

   d. Complete the statement.

   To make this shade of lilac paint, mix _____ cups of blue paint for every _____ cups of red.

# Unit 5, Lesson 3: Exploring Rates

### Benchmark

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

### Learning Targets

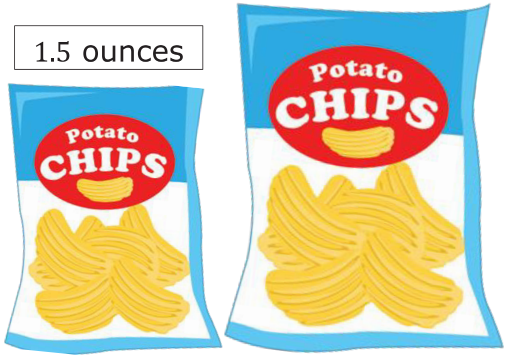
- ☐ I can use ratios and rates to describe relationships between quantities.
- ☐ I can write a ratio as a unit rate.

## 5.3.1 Warm-Up: Would You Rather?

1. Would you rather share the small bag of chips with 1 friend or the large bag of chips with 8 friends? Explain your reasoning using mathematics.

8 ounces

1.5 ounces




## 5.3.2 Collaboration: Expressing Ratios in Different Ways

Ratios are often expressed in the real world as an amount of "something per something." For example, in setting the world record, Joey Chestnut ate 7.4 hot dogs per minute.

Discuss each of the following scenarios with your partner and work together to answer the questions.

1. A recipe for marinara sauce requires 56 ounces (oz) of whole peeled tomatoes. The recipe yields 8 servings of marinara sauce.

   a. What is the ratio of tomatoes to sauce?

   b. Complete the tape diagram to represent the ratio of tomatoes to sauce.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Tomatoes (ounces) | | | | | | | | |
| Sauce (servings) | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

   c. How many ounces of whole peeled tomatoes are in 1 serving of marinara sauce?

   d. Complete the ratio.

$$\frac{\_\_\_\_\_\_\text{oz of tomatoes}}{1\text{ serving}}$$

2. In 2012, there were 982 farms in Hardee County, Florida. The total amount of farmland was 273,978 acres.

   a. Write the ratio of acres : farms.

   b. Complete the statement.

   The total amount of farmland in Hardee County is split up among all the farms, so to determine the average number of acres for 1 farm,

   - ○ add 982 to 273,978.
   - ○ subtract 982 from 273,978.
   - ○ multiply 273,978 by 982.
   - ○ divide 273,978 by 982.

   c. Complete the ratio.

   $$\frac{\_\_\_\_\_\_\_\_\text{acres}}{1\text{ farm}}$$

   d. Discuss with your partner whether you think this number of acres per 1 farm means that every farm in Hardee County was that size in 2012. Then, write a brief summary of your reasoning.

## 5.3.3 Guided Instruction: Understanding Rates

Each of the ratios discussed so far in this lesson are also ***rates***.

A ***rate*** is a ratio that compares two quantities of different units.

A ratio is a comparison of two numbers, so all rates can also be expressed using ratios. Rates are often used to express how long it takes to do something.

1. Consider a familiar scenario involving a rate.

   Tyler ran 2 miles in 30 minutes.

   a. Draw a tape diagram for the ratio that represents Tyler's run.

   b. Complete the ratios. Then, complete each statement to interpret the ratios.

   $$\frac{\_\_\_\_\_\_\_\_\text{miles}}{1\text{ minute}} \qquad \frac{\_\_\_\_\_\_\_\_\text{minutes}}{1\text{ mile}}$$

   Tyler runs _______ miles for every 1 minute.

   Tyler runs _______ minutes for every 1 mile.

c. Discuss with your partner how to use one of the rates to determine how many miles Tyler runs in an hour. Then, complete the statement.

Tyler runs _______ miles per hour.

Each of the rates used to describe Tyler's run is a ***unit rate***.

A ***unit rate*** is a ratio comparing a number of units of one quantity to 1 unit of a second quantity.

## 5.3.4 Guided Practice: Using Ratios and Rates

1. A nutritional label for raspberry ice cream is shown.

**Nutrition Facts**

3 servings per container

| **Serving size** | | | **2/3 Cup (123g)** |
|---|---|---|---|
| | | **Per Container** | |
| **Calories** | | **1000** | |
| | | | **% Daily Value*** |
| **Total Fat** | | 58g | **74%** |
| Saturated Fat | | 36g | **180%** |
| *Trans* Fat | | 1.5g | |
| **Cholesterol** | | 385mg | **128%** |
| **Sodium** | | 550mg | **24%** |
| **Total Carbohydrate** | | 106g | **39%** |
| Dietary Fiber | | 0g | **0%** |
| Total Sugars | | 88g | |
| Includes Added Sugars | | 72g | **144%** |
| **Protein** | | 16g | |
| Vitamin D | | 3mcg | 15% |
| Calcium | | 405mg | 30% |
| Iron | | 2mg | 10% |
| Potassium | | 532mg | 10% |

*The % Daily Value tells you how much a nutrient in a serving of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.

a. What is the ratio of calories to servings?

b. What is the ratio of total carbohydrates in the container to the number of calories in the container?

c. What is the ratio of added sugars to the number of servings?

d. Complete the unit rate to express the number of added sugars per serving.

$$\frac{\_\_\_\_\_\_\text{ grams}}{1\text{ serving}}$$

## 5.3.5 Your Turn!

1. Tracy McGrady is a former NBA basketball player from Bartow, Florida. He played in $938$ games in the NBA, scoring a total of $18{,}381$ points. Complete the ratio to express the number of points he scored per game as a unit rate. Round the number of points to the nearest tenth.

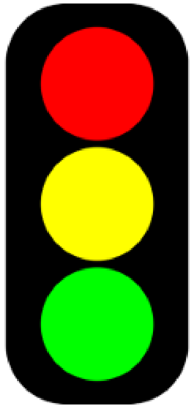
$$\frac{\underline{\qquad}\ \text{points}}{1\ \text{game}}$$

2. The ruby-throated hummingbird is the most common hummingbird found in Florida. This bird can flap its wings $1{,}590$ times in $30$ seconds. Complete the ratio to express how fast the ruby-throated hummingbird flaps its wings per second.

____ wing flaps : $1$ second

## 5.3.6 Wrap-Up: Reflection

1. Complete the stoplight reflection to share how you are feeling about the concepts learned in today's lesson.



I still feel stuck about how to …

I'm starting to get the hang of …

I feel confident and ready to go with …